PARISHRAM 2.0

CLASS 12TH 2026

Inverse Trigonometric Functions

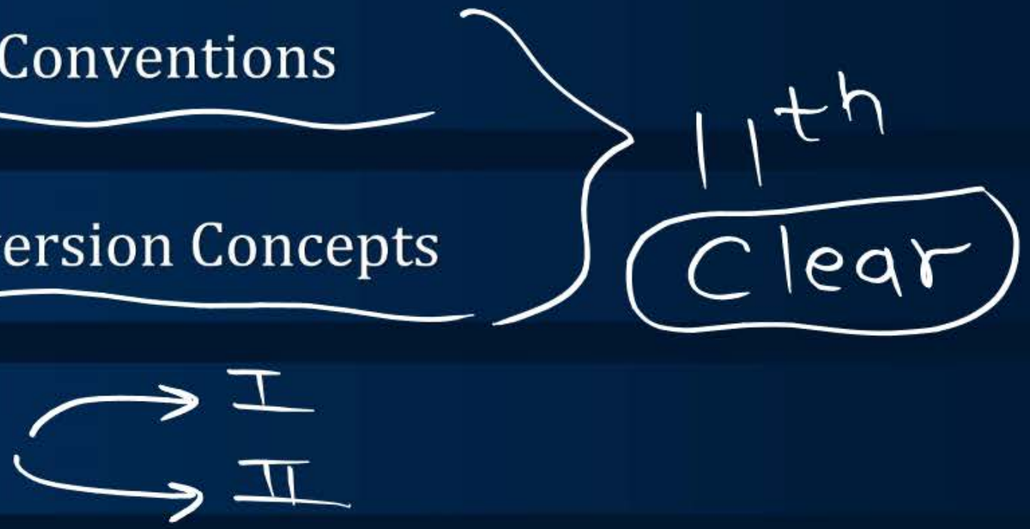
MATHS | LECTURE-2

BY-MANDAR SIR



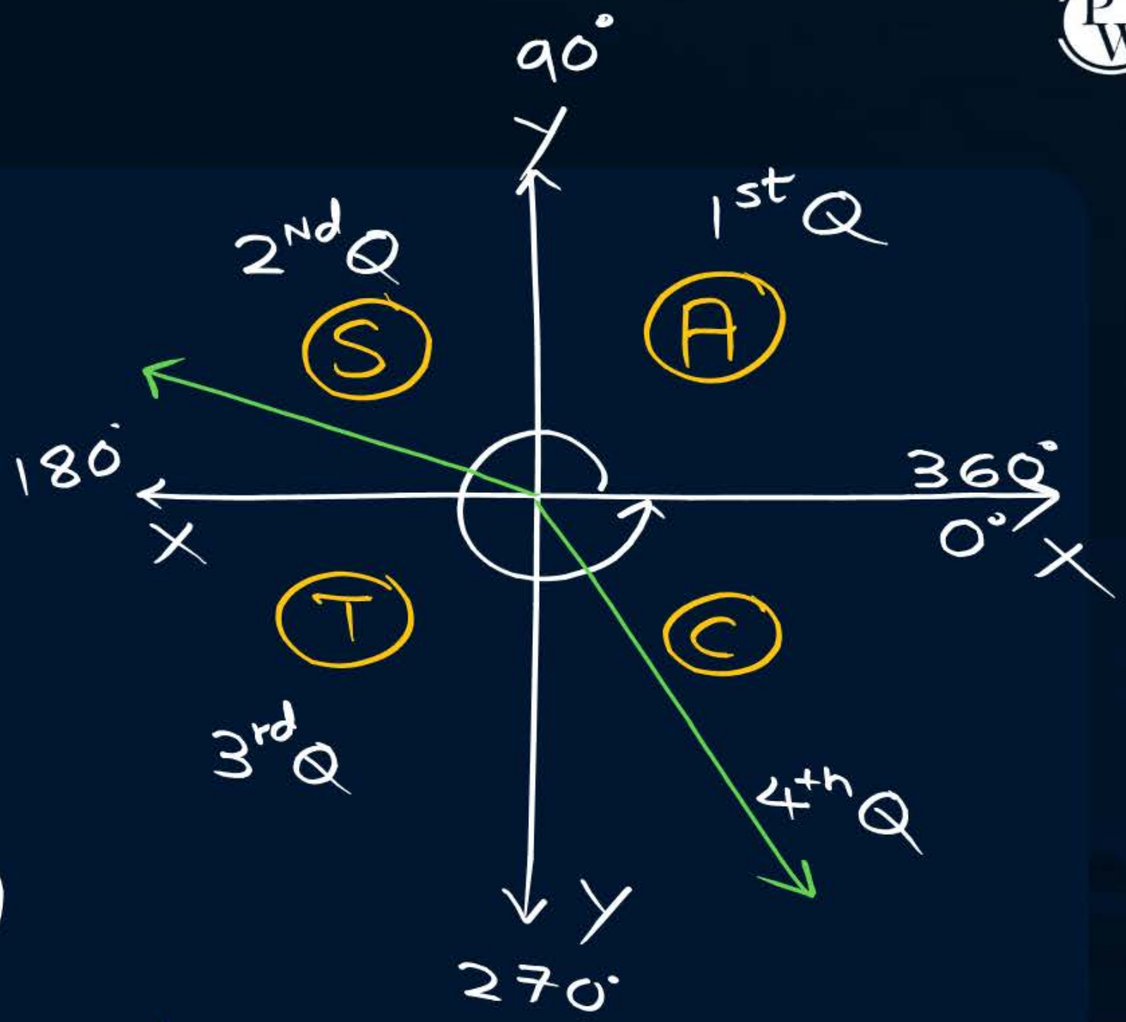
TOPICS TO BE COVERED

1. Trigo Sign Conventions
2. Trigo Conversion Concepts
3. Properties
4. Questions Based on Properties



Trigo Sign Conventions

ASTC
→ Add Sugar to Coffee
→ After School to College



A → All Trigo values → +ve

S → sin, cosec → +ve; Rest → -ve

T → tan, cot → +ve; Rest → -ve

C → cos, sec → +ve; Rest → -ve

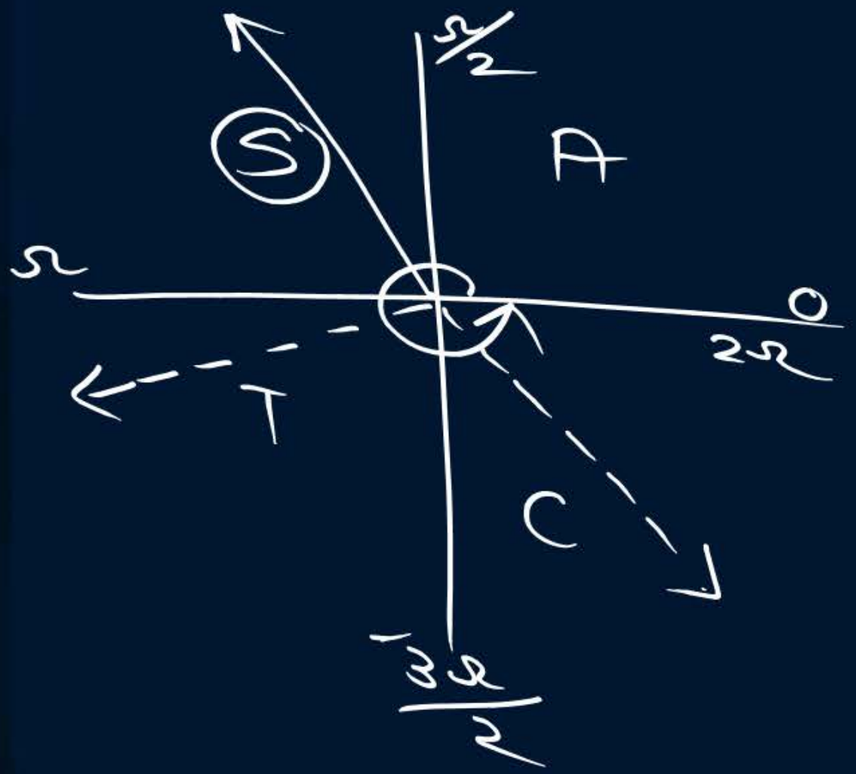
tan 170° → negative
cos 300° → positive

Trigo Conversion

sin ↔ cos

tan ↔ cot

cosec ↔ sec



* $\frac{\pi}{2}$ → multiple ✓

$\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$ ....

$\frac{2\pi}{2} \rightarrow \pi$

$\sin\left(\frac{\pi}{2}+\theta\right) = +\cos\theta$

$\tan\left(\frac{\pi}{2}+\theta\right) = -\cot\theta$

$\sin\left(\frac{3\pi}{2}+\theta\right) = -\cos\theta$

* π → Trigo convert X

π → Multiple

→ $\pi, 2\pi, 3\pi, 4\pi, \ldots$

$\cos(\pi+\theta) = -\cos\theta$

Comment

- $\tan\left(\frac{\pi}{2}+\theta\right)$
- $\cot(3\pi+\theta)$
- $\sin(2\pi-\theta)$
- $\cot\left(\frac{5\pi}{2}+\theta\right)$
- $\tan(4\pi-\theta)$

Properties Set 1

(i) $\sin^{-1}(\sin\theta) = \theta$, for all $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(ii) $\cos^{-1}(\cos\theta) = \theta$, for all $\theta \in [0, \pi]$

(iii) $\tan^{-1}(\tan\theta) = \theta$, for all $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(iv) $\operatorname{cosec}^{-1}(\operatorname{cosec}\theta) = \theta$, for all $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \neq 0$

(v) $\sec^{-1}(\sec\theta) = \theta$, for all $\theta \in [0, \pi]$, $\theta \neq \frac{\pi}{2}$

$\cot^{-1}(\cot\theta) = \theta$

(vi) $\cot^{-1}(\cot\theta)$, for all $\theta \in (0, \pi)$

$\frac{4\pi}{5} = \frac{4\times180}{5} = 4\times36 = 144^\circ$

θ

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right) =$

$\theta \to -\frac{\pi}{2}, \frac{\pi}{2}$ } Beech

$\frac{2\pi}{3} = 60^\circ$

$\sin^{-1}\left(\sin\left(\frac{4\pi}{5}\right)\right)$

$\frac{4\pi}{5} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ X

Properties Set 2

(i) $\sin(\sin^{-1}x) = x$, for all $x \in [-1, 1]$

(ii) $\cos(\cos^{-1}x) = x$, for all $x \in [-1, 1]$

(iii) $\tan(\tan^{-1}x) = x$, for all $x \in R$

(iv) $\text{cosec}(\text{cosec}^{-1}x) = x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

(v) $\sec(\sec^{-1}x) = x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

(vi) $\cot(\cot^{-1}x) = x$, for all $x \in R$

POLL QUESTION [NCERT Exemplar]

$\sin 30^\circ = \frac{O}{H} = \frac{1}{2}$

The value of the expression $2\sec^{-1}2 + \sin^{-1}\left(\frac{1}{2}\right)$ is

1. $\pi/6$
2. $5\pi/6$
3. $7\pi/6$
4. 1

$2\sec^{-1}2 + \sin^{-1}\left(\frac{1}{2}\right)$

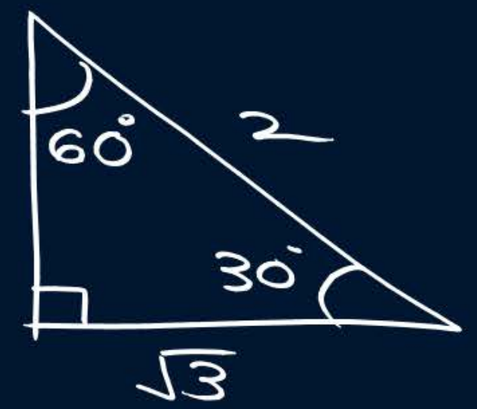
$2\sec^{-1}(\sec 60^\circ) + \sin^{-1}(\sin 30^\circ)$

$2(60^\circ) + (30^\circ)$

$2 \times \left(\frac{\pi}{3}\right) + \left(\frac{\pi}{6}\right)$

$\frac{2\times 2\pi}{2\times 3} + \frac{\pi}{6} \rightarrow \frac{4\pi}{6} + \frac{\pi}{6}$

$= \frac{5\pi}{6}$



$\sec 60^\circ = \frac{H}{B}$

$\sec 60^\circ = \frac{2}{1}$

POLL QUESTION [CBSE, 2020]

The value of $\sin^{-1}\left(\cos\frac{3\pi}{5}\right)$ is

1. $\pi/10$
2. $3\pi/5$
3. $-\pi/10$
4. $-3\pi/5$

$18^\circ = \frac{\pi}{10}$



$\sin^{-1}\left(\cos\frac{3\pi}{5}\right)$

$\sin^{-1}\left(\cos 108^\circ\right)$

$\sin^{-1}\left(\cos(90+18)\right)$

$\sin^{-1}\left(\cos\left(\frac{\pi}{2}+\frac{\pi}{10}\right)\right)$

$\sin^{-1}\left(-\sin\frac{\pi}{10}\right) = -\sin^{-1}\left(\sin\frac{\pi}{10}\right)$

$= -\frac{\pi}{10}$

$\sin^{-1}(-x) = -\sin^{-1}x$

$\frac{3\pi}{5} = \frac{3\times 180}{5} = 108^\circ$

$\frac{\pi}{10} \to -\frac{\pi}{2} \to \frac{\pi}{2}$

QUESTION

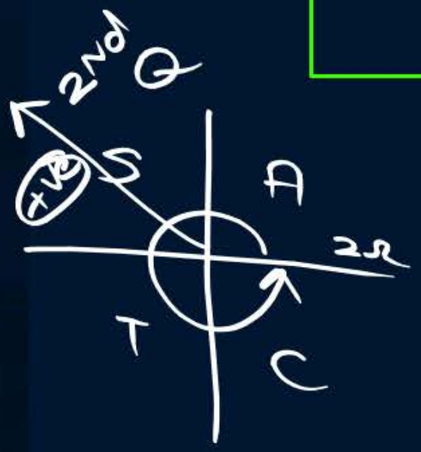
Find the principal value of $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$.

$$\sin^{-1}(\sin x) = x$$

$$x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\frac{3\pi}{5} = \frac{3\times 180^{36}}{5_1} = 108^\circ$$

$-\frac{\pi}{2}$ (-90°) $\quad$ $\frac{\pi}{2}$ (90°)



$$\sin^{-1}(\sin 108^\circ)$$

$$\sin^{-1}(\sin(180-72))$$

$$\sin^{-1}\left(\sin\left(\pi-\frac{2\pi}{5}\right)\right)$$

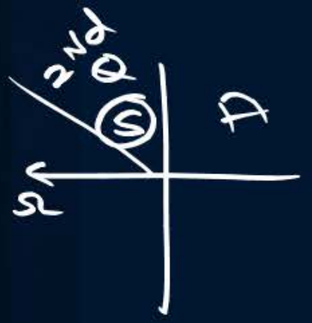
$$\sin^{-1}\left(+\sin\frac{2\pi}{5}\right) = \frac{2\pi}{5}$$

$-\pi/2$ ↩ $\pi/2$

QUESTION

$\cos^{-1}(\cos x) = x \quad x \in [0, \pi]$

Evaluate : $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$



$\sin^{-1}(\sin 135^\circ) + \cos^{-1}(\cos 180^\circ) + \tan^{-1}(1)$

$\sin^{-1}(\sin(180-45)) + 180^\circ + \tan^{-1}(\tan 45^\circ)$

$\sin^{-1}\left(\sin\left(\pi - \frac{\pi}{4}\right)\right) + \pi + \tan^{-1}\left(\tan\frac{\pi}{4}\right)$

$\sin^{-1}\left(+\sin\frac{\pi}{4}\right) + \pi + \frac{\pi}{4}$

$\frac{\pi}{4} + \pi + \frac{\pi}{4} = \frac{\pi}{4} + \frac{4\pi}{4} + \frac{\pi}{4}$

$= \frac{\pi + 4\pi + \pi}{4} = \frac{6\pi}{4} = \boxed{\frac{3\pi}{2}}$

$\frac{3\pi}{4}$

$\frac{3\times 180}{4}$

$3 \times 45^\circ$

135°

QUESTION [NCERT Exemplar]

Evaluate : $\sin^{-1}\left[\cos\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right]$

$\sin^{-1}\left[\cos\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right]$

$\sin^{-1}\left[\cos\left(\sin^{-1}(\sin 60^\circ)\right)\right]$

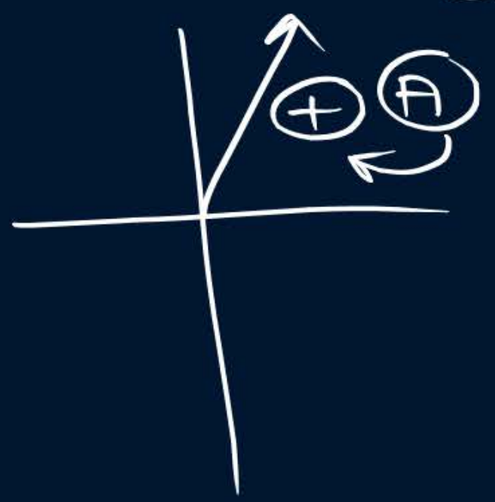
$\sin^{-1}\left[\cos 60^\circ\right]$

$\sin^{-1}\left[\cos(90-30)^\circ\right]$

$\sin^{-1}\left[\cos\left(\frac{\pi}{2}-\frac{\pi}{6}\right)\right] = \sin^{-1}\left(+\sin\frac{\pi}{6}\right)$

$= \boxed{\frac{\pi}{6}}$

$\sin 60^\circ = \frac{\sqrt{3}}{2}$




"Yeh Azadi humko tofe mein nahi mili, iske peeche laakho shaheedon ka khoon hai.
Tiranga sirf ek jhanda nahi, yeh hamari shaan, hamari jaan hai.
Jab tak humari saansen chalti hain, is mitti ka karz hum chukate rahenge.
Vande Mataram - yeh sirf shabd nahi, ek amar kasam hai."

THANK YOU